



Learning neural networks for text classification by exploiting label relations

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Abstract

Text classification has been a fundamental problem in the realm of Natural Language Processing (NLP), and a variety of approaches have been proposed with the development of deep learning. Despite recent progress, most existing approaches deal with the problem of multi-class text classification in a flat way, assuming that text labels are semantically independent. This assumption doesn't always hold in realistic settings, since there usually exist hierarchical or within-layer dependencies in the latent label space, especially when we consider larger label sets. In this paper, we propose a label clustering algorithm to exploit the underlying structure of label relations, and express the stacked concept relationships in the form of a two-layer label space. Next, we present two different neural network structures to capture inter-layer and intra-layer label relations. The first model HSNN organizes a group of local classifiers in a hierarchical way according to the exploited label space, while the second model LSNN takes advantages of text representations in different granularity levels and the bidirectional inferences with recurrent connections to make predictions. Finally, we evaluate our methods on three benchmark datasets. The results empirically demonstrate that both models are capable of leveraging the exploited label relations to improve text classification performance.

Keywords Text classification · Label relations · Hierarchical agglomerative clustering

1 Introduction

Text classification has received increasing attention because of its significant value in assigning predefined labels to a large bulk of documents automatically. Besides, it serves as a building block for a variety of NLP applications, such as information retrieval, question answering, reading comprehension and so on. Although notable advancements have

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been made in recent years [14, 15, 17], most work in the literature usually assume labels are mutually exclusive in multi-class text classification and ignore the rich structure of label relations.

Here we take label relations as the pairwise distance between labels in a semantically meaningful space. Take the following case as an example. A document talking about “cat” may be semantically similar to the document talking about “dog”, but much different from the topic of “email”. The semantic relations between labels can exert an influence on the classifier. Given such a 3-class classification task, if the classifier learns to take advantages of the underlying label relations and pays more attention to the distinction between the closely correlated labels “cat” and “dog” rather than treat all labels equally, it could extract more specific features and subsequently improve its performance.

In this paper, we study how to improve text classification performance in a deep learning framework by exploiting label relations. We shed light on the exploitation of label relations with clustering techniques, and give an in-depth technical description of the strategies. Then we present two different networks, including the Hierarchically Structured Neural Network (HSNN) which leverages our generated label space in a hierarchical structure, and the Long-Short Term Memory Neural Network in a two-layer Label Space (LSNN) which takes text representations in different granularity levels as inputs. While both models aim to exploit label relations, HSNN utilizes different groups of local information in a top-down way and LSNN unifies cluster-specific and label-specific information in a parallel approach.

The contribution of our work can be summarized as follows: (1) We present a cluster-based algorithm to exploit label relations without human supervision. (2) We present two different networks termed as HSNN and LSNN to improve the performance of a vanilla text classification model through label relations. (3) We perform extensive experiments to demonstrate the rationality and validity of our label clustering algorithm, as well as evaluate our proposed models on multiple benchmark datasets in comparison with baselines. The experimental results are promising.

2 Related work

A significant number of studies have investigated various methods to solve multi-class or multi-label classification problems with the help of label relations. Some of these studies organize the class labels manually into a tree or a directed acyclic graph, and then propose hierarchical classification methods to make use of label relations. Top-down approaches are typically employed along the hierarchy in the way of a local classifier per node, a local classifier per parent node or a local classifier per level. See [25] for surveys. While the hierarchical or within-layer dependencies of label relations are mainly obtained from human comprehension here which can be time-consuming and error-prone, a family of studies capitalize on structured label prediction with external knowledge. One of the most important knowledge sources is WordNet [21] category taxonomy, and it has been widely used to model label relations in concept layers [11–13, 22].

Label relations in the real world are far more complicated than the simple assumption that the children are always mutually exclusive in the hierarchy [5]. Therefore, the authors introduce HEX graphs in [5] to capture the complex semantic relations between different labels including mutual exclusion, overlap and subsumption for the study of object classifi-

cation, and in [8] the classification model based on HEX graphs is further extended to the Ising model to allow for soft or probabilistic relations between labels. While most previous work exploit label relations manually or with external knowledge, we propose a novel way with clustering techniques to exploit label relations automatically in the latent semantic space.

In recent years, label relations exploitation have been integrated into the deep learning framework, especially in the field of Computer Vision (CV). For example, a sequential inference method modeled by Recurrent Neural Networks (RNN) is proposed in [6] to learn label relations for group activity recognition. In [11], a structured inference neural network is presented to classify images in the layered label space with external label relations. Besides, exploiting label relations is closely related to zero-shot learning or few-shot learning. In [18], the authors propose a novel model which incorporates knowledge graphs to describe label relations for multi-label zero-shot learning. In [28], a GCN-based approach using the semantic embeddings of a category and the relationships between categories is proposed to learn a visual classifier under the setting of zero-shot recognition.

In NLP, there has been a remarkable series of work for text classification [10, 14, 15, 17, 29] with deep learning recently, whereas none of them takes label relations into consideration. Most prior work on exploiting label relations are concerned with image classification or object classification in CV. It is worth mentioning that label relations can not be leveraged in the same way for CV and NLP. For example, “cat” and “dog” are mutually exclusive in image classification because they are two kinds of animals totally different in appearance. However, in NLP they often appear in a similar textual context and their semantics are overlapped. We specially aim to solve the problem of exploiting label relations for text classification in this paper, which unifies text classification, deep learning and label relations exploitation into a single framework.

3 Framework

The overall framework is illustrated in Fig. 1. Given a set of text labels, we first transform them into distributed vectors through word embeddings, and exploit structured label

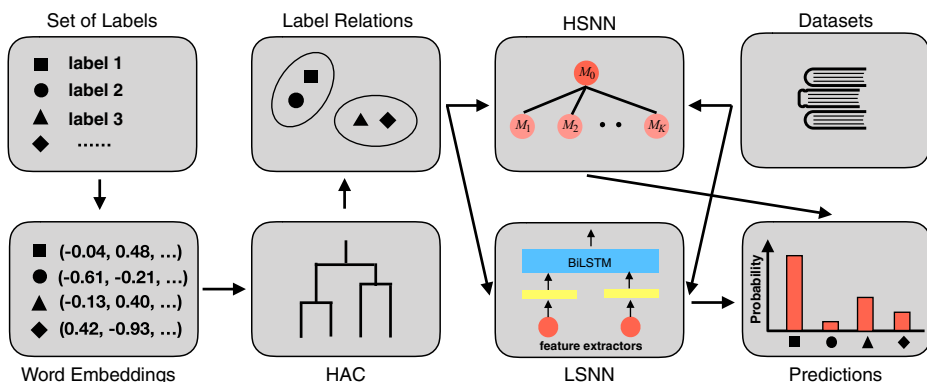


Fig. 1 Framework

relations with clustering techniques. After that, we use HSNN and LSNN to capture inter-layer and intra-layer label relations to perform text classification. We will go through the framework in detail in the following sections.

3.1 Label clustering

Now we describe how to exploit label relations with clustering techniques. Firstly we introduce the formalism of the representation for our structured label relations as follows:

Definition 1 *Given a set of N labels and a distance measure function $\mathcal{F}$, label relations are expressed in the form of a hierarchical two-layer label space $\mathcal{O} = (\mathcal{C}, \mathcal{L})_{\mathcal{F}}$ with clustering techniques, where $\mathcal{C}$ and $\mathcal{L}$ represent the first and second concept layers respectively. The first concept layer consists of K clusters $\mathcal{C} = \sum_{i=1}^K c_i$, and the second concept layer consists of N labels $\mathcal{L} = \sum_{i=1}^N l_i$. The following conditions should be satisfied: $c_i = \sum_j^{j+size(c_j)} l_j$, $c_i \cap c_{i'} = \emptyset$ and $N = \sum_{i=1}^K size(c_i)$, where l_j is a particular label in the cluster c_i .*

Even though achieving the optimal clustering results is an NP-hard problem which is computationally difficult, it suffices to obtain an approximately optimal solution in a relatively efficient way. Traditional clustering algorithms such as K-means can provide a heuristic solution to find a local optimum. However, the number of clusters K has to be determined before the partition, which can be problematic for the exploitation of label relations. Here we adopt a simple yet effective algorithm, Hierarchical Agglomerative Clustering (HAC). The algorithm treats every label as a singleton cluster, and then agglomerates the two most similar clusters successively according to one of the similarity measures (e.g., complete-link clustering, single linkage or average linkage). The process will not stop until all labels have been merged into one large singleton cluster. We combine it with the pre-trained word embeddings popular in NLP to exploit label relations.

In order to form the two-layer label space, we want to determine the best cluster configuration in a flat clustering way. A quality function is required to select the cutting point in the resulting *dendrogram*, which is a typical visualization of the process when HAC successively merges clusters at each level.

For the purpose of measuring cluster similarities, it is straightforward to compute the pairwise semantic distance between any two labels in the distance measure function $\mathcal{F}$. There has been plenty of methods proposed in the literature to estimate the semantic similarity [4, 9, 20]. We use the Euclidean distance of distributed vector representations [20] as $\mathcal{F}$. Word embeddings are usually pre-trained on an extremely large corpus. Each floating point in a word vector contains a dimension of the word's meaning and semantically similar words can be mapped into proximate vector space. Hence, it is appropriate to model label relations in NLP tasks through the exploitation of word embeddings.

All the labels will be transformed into distributed representations for further clustering. If a label contains more than one word, we ignore meaningless ones such as stop words. Necessary lemmatizations or expansions have to be performed to make sure the availability of word representations in the pre-trained word embeddings. Then we average the value of word embeddings belonging to a certain label into distributed label representations. The hypothesis here is that a non-obvious degree of language understanding can be achieved with basic mathematical operations on word embeddings [20], for example, $\text{vec}(\text{"king"})$

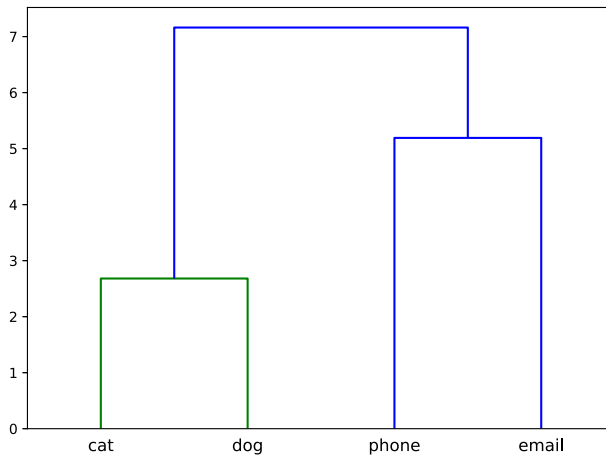


Fig. 2 The dendrogram generated by HAC with example labels

- $\text{vec}(\text{"man"}) + \text{vec}(\text{"women"})$ is close to $\text{vec}(\text{"queen"})$. After distributed label representations have been acquired, the clustering algorithm is able to exploit the semantic label relationships.

To demonstrate the process of label clustering, we display an example of the generated dendrogram in Fig. 2. The vertical axis represents the similarity distance between clusters with average linkage algorithm and the horizontal axis represents text labels.

Even though our structured label relations consist of only two layers, HAC can be performed recursively when the size of some clusters are too large and it is necessary to exploit deeper label relations to generate a multi-layer structure from coarse to more fine-grained levels. The reasons for our two-layer label space instead of a multi-layer structure are twofold: (1) Most real-world text classification tasks are concerned with a limited number of labels, and two consecutive concept layers are enough to model label relations elaborately. (2) A classification result will be propagated along the concept hierarchy. It is essential to produce a practical trade-off between the exploitation of label relations and the propagation of errors originating from a certain level in the long hierarchy path.

3.2 HSNN: Hierarchically structured neural network

Inspired by the idea of hierarchical multi-label classification with local neural networks [1], it makes sense to train a group of local classifiers in a top-down way to perform multi-class classification. The network architecture of HSNN is presented in Fig. 3. Inspired by the exploited label relations, we can organize the hierarchical structure according to the two-layer label space, where in the first layer the decision boundary is robust for coherent semantic spaces of the documents belonging to different clusters, and in the second layer classifiers are trained to learn fine-grained features within each cluster.

Given a two-layer label space $\mathcal{O} = (\mathcal{C}, \mathcal{L})_{\mathcal{F}}$ and the base learner $\mathcal{M}$, a local classifier $\mathcal{M}_0$ is trained to classify inputs into K clusters first, and K more local classifiers $\mathcal{M}_1 \sim \mathcal{M}_K$ are trained in turn to perform text classification for all the labels in each cluster c_j . The motivation for this model is that hierarchical relations are taken into account using a local information perspective [25]. Here local information is the subset of data with respect to each parent node in the two-layer label space required to train a local classifier.

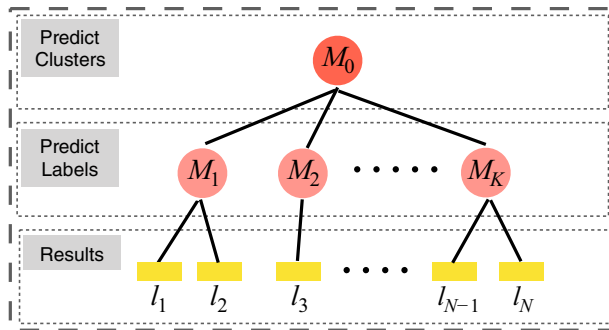


Fig. 3 The network architecture of HSNN

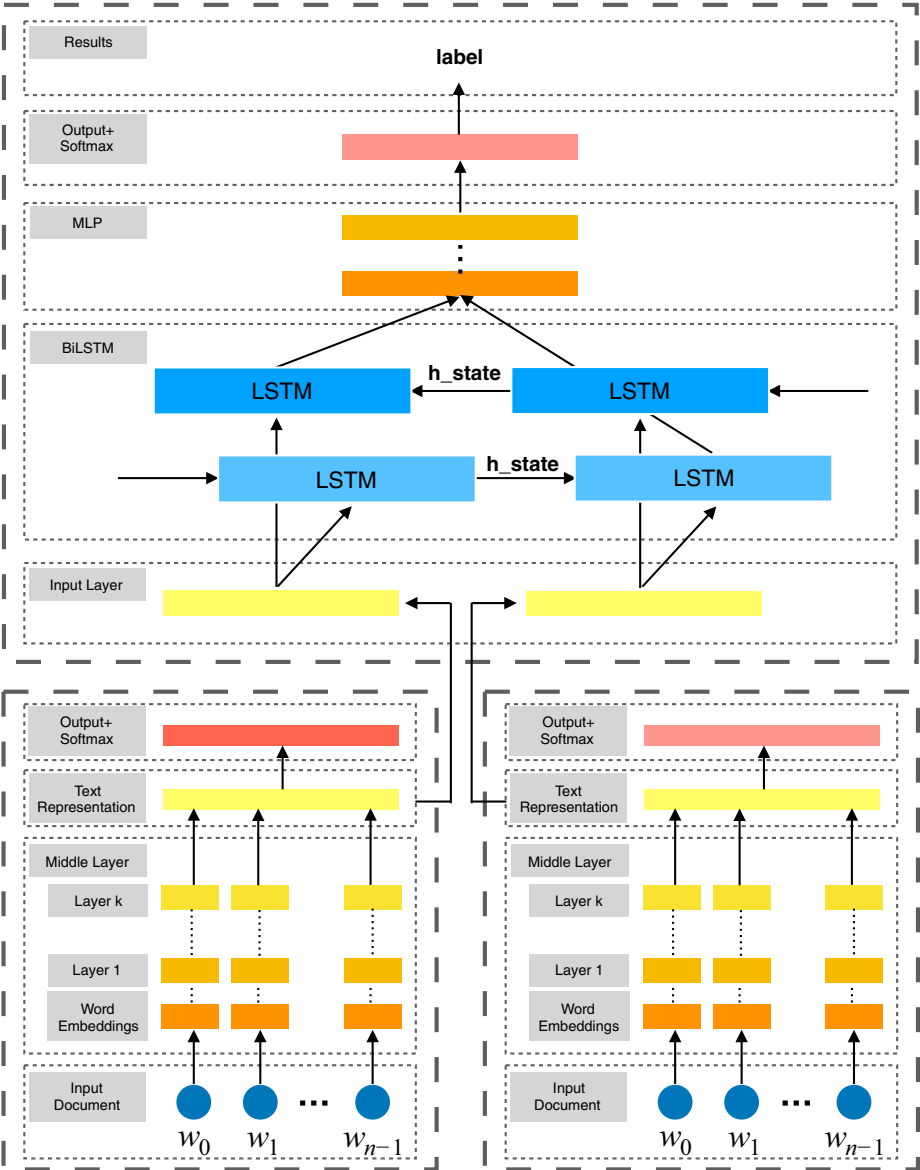
In HSNN, inter-layer label relations are captured by the hierarchical structure, which means the ensemble of local classifiers organized by the two-layer label space will predict the category of inputs from coarse to fine-grained in a progressive way. More concretely, the local classifier $\mathcal{M}_0$ learns high-level features in the first level, and separates inputs into cluster-specific predictions to narrow down the predictions at the next level, while the other K local classifiers learn low-level features for second-level label-specific predictions. On the other hand, intra-layer label relations are captured by each local classifier. In general, a large number of classes lead to difficulties to discriminate the characteristics among them and make the correct prediction. With the help of the exploited stacked label structure, the original label space is decomposed into several smaller sub-spaces, and the documents belonging to each sub-space share more coherent semantics. Consequently, the local classifier for each sub-space can focus on specific intra-layer label relations with respect to local information.

The choice of the base learner $\mathcal{M}$ is arbitrary as long as it is applicable for text classification. In fact, the goal of HSNN is to improve the predictive performance of $\mathcal{M}$ in the cascaded structure, and the accuracy gains come from the exploitation of label relations in the two-layer label space.

3.3 LSNN: Long-short term memory neural network in a two-layer label space

In the architecture of HSNN, label relations are leveraged explicitly to help associate local classifiers to the hierarchical structure. In this section, we propose another deep learning model LSNN to learn constraints between two consecutive layers implicitly. The architecture of LSNN is inspired by the BINN model in [11], where the authors improve image classification performance with a novel stacked label prediction neural network. The network architecture of LSNN is shown in Fig. 4. Recent development on LSTM include [26, 27]. In fact, the component of LSTM inside LSNN can be safely replaced by RNN or GRU [3] to capture label relations.

It has been well recognized that the learned text representations can be used as input features in the downstream NLP tasks [7, 19]. In the LSNN model, we train two neural network classifiers in advance as feature extractors to embed textual inputs and generate text representations for LSTM inputs. The basic structures of the two feature extractors keep the same except for the output layer. In more details, they are trained to classify inputs into K clusters and N labels respectively in the two-layer label space, and the resulting text representations in terms of different granularity levels (cluster-specific or label-specific)



the feature extractor for cluster-specific text representations **the feature extractor for label-specific text representations**

Fig. 4 The network architecture of LSNN

are fed into consecutive timesteps in LSTM. The output of LSTM in the last timestep is followed by a multiple layer perceptron, and label-specific probabilities are obtained from softmax.

According to the design principle of LSNN, we can use various architectures for the feature extractor, including deep learning models typically designed for text classification (*e.g.*, fastText [14] and TextCNN [31]), or pre-trained language models (*e.g.*, BERT [7]). Even though our exploited label relations consist of only two layers, it can be easily extended to a multi-layer structure when necessary as remark earlier. Under these circumstances, more feature extractors have to be trained for each granularity level, and the LSTM structure employed in our LSNN can contain more than two timesteps where each individual unit corresponds to a certain concept layer in the hierarchical structure.

More formally, the label relations capturing process is described in (1), where C_t , $\tilde{C}_t$, i_t , f_t , o_t denote the internal state of the memory cells, new candidate values, input gate, forget gate and output gate in LSTM. h_t is the hidden state at timestep t . Each cell captures intra-layer label relations through the gate structures inside an individual LSTM unit, while the recurrent connections between consecutive timesteps capture inter-layer label relations by means of the hidden state.

$$\begin{aligned} C_t &= f_t * C_{t-1} + i_t * \tilde{C}_t \\ h_t &= o_t * \tanh(C_t) \end{aligned} \quad (1)$$

The component of LSTM in our model is in the form of a bidirectional RNN. The process of bidirectional inferences is akin to aggregating inter-layer label relations in both top-down and bottom-up propagations in the hierarchy of label relations. The outputs in both directions are concatenated together into final activations to make predictions, as shown below:

$$\begin{aligned} \vec{h}_t &= LSTM_f(x_t, \vec{h}_{t-1}) \\ \overleftarrow{h}_t &= LSTM_b(x_t, \overleftarrow{h}_{t+1}) \\ h_t &= f(\vec{h}_t, \overleftarrow{h}_t) \end{aligned} \quad (2)$$

We have adopted the softmax to generate the probability distribution over all labels and we want to assign one predefined label to each document for multi-class text classification. To update the model parameters in LSNN using backpropagation algorithm, we minimize the negative log likelihood loss between the predicted label-specific probabilities and the real label.

4 Experiments

We have performed four sets of experiments in this section on three text classification datasets for they have a relatively large label set to exploit the inner relations: (1) 20 Newsgroups¹: A popular text classification dataset partitioned (almost) evenly across 20 different newsgroups. (2) DBpedia [30]: An ontology dataset consisting of 14 non-overlapping classes from DBpedia 2014. (3) Yahoo! Answers [30]: A corpus with question-answer pairs from the Yahoo! Webscope program. The statistics of the datasets are shown in Table 1. We firstly exploit label relations with our proposed label clustering algorithm. Then we conduct extensive experiments to evaluate how different choices for the value of K in HAC would affect the text classification performance of HSN. We analyze the effects of different granularity levels of text representations on the performance of LSNN subsequently. Lastly we report the experimental results of HSN and LSNN in comparison with different baselines

¹<http://qwone.com/~jason/20Newsgroups/>

Table 1 Statistics of the three text classification datasets in our experiments

Datasets	Labels	Train Samples	Test Samples
20 Newsgroups	20	15,043	3,778
DBpedia	14	560,000	70000
Yahoo!Answers	10	1,400,000	60,000

to demonstrate the effectiveness of the overall framework. All the results are reported in accuracy on the test set.

4.1 Experiments I: Exploiting label relations through the label clustering algorithm

This experiment shows the exploitation of label relations with clustering techniques. We use the pre-trained 100-dimensional GloVe [23] vectors to embed labels into distributed vectors. The average linkage algorithm is employed to measure the similarity between two clusters. In order to get a partition of disjoint clusters in HAC and determine the approximately optimal number of clusters, we compute the silhouette [24] as the quality function to measure every possible cutting point of all levels in the dendrogram. The silhouette for an object $s(i)$ can be calculated with $a(i)$ which denotes the average distance between i and other objects in the same cluster, and $b(i)$ which denotes the smallest average distance of i and all objects in any other cluster, as shown below:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (3)$$

The overall quality S of the resulting partition can be measured in the way of the average value of $s(i)$ for every object in the dataset, as illustrated in (4). A larger S means a better cluster configuration.

$$S = \frac{1}{n} \sum_{i=1}^n s(i), \quad -1 \leq S \leq 1 \quad (4)$$

The relationships between the average silhouette value S and the number of clusters K are depicted in Fig. 5. We also mark the near-optimal number K with respect to the largest S in the picture. Since some labels in 20News are very closely related to each other in semantic and they even share the same word, the results for 20News demonstrate higher S .

We also zoom in on the detailed silhouette value for every label in 20News for illustration purposes. The results are displayed in Fig. 6. Note that the silhouette value equals to zero if there is only one object in the cluster (e.g., c4 and c7 in Fig. 6).

4.2 Experiments II: Effects of different K in HAC on HSNN

We perform experiments to evaluate the relationships between the number of K in HAC and the performance of HSNN. More concretely, we exploit the dendrogram of HAC in detail and vary the number of clusters K from 1 to 10 for different implementations of HSNN on 20News.

We choose the text classification model fastText [14] as the local classifier to build HSNN. Our local reproduction of fastText averages the randomly initialized word embeddings into a text representation, followed by a linear layer of 64 hidden units and trained by the Adam optimizer [16]. We save every local classifier with the best test accuracy during 20 epochs to report the overall performance of HSNN.

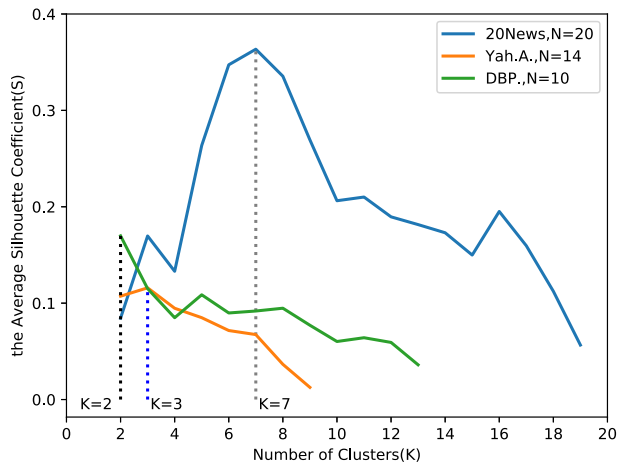


Fig. 5 The average silhouette value when varying K in HAC

Figure 7 illustrates the experimental results. When $K=1$, the model can be regarded as a baseline which consists of only one classifier to deal with all the classes in a flat way and label relations are not taken into consideration. When $K=2\sim 10$, label relations are involved in the hierarchical classification. An interesting observation is that the test accuracy reaches a local maximum when $K=3$ (compared to $K=2$ or 4), which is consistent with the results in Fig. 5 where the average silhouette value for $K=3$ is the largest in the same neighborhood. We can see that the changes in curves are generally similar for Figs. 5 and 7. On the other hand, the test accuracy increases moderately as we increase the number of clusters from $K=4$ in HSNN, which demonstrates the benefits of exploiting label relations. When $K=7$, the test

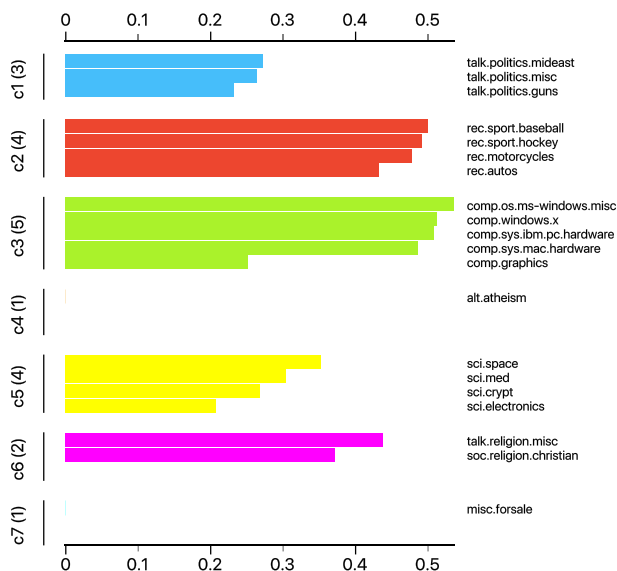


Fig. 6 The silhouette value for 20News

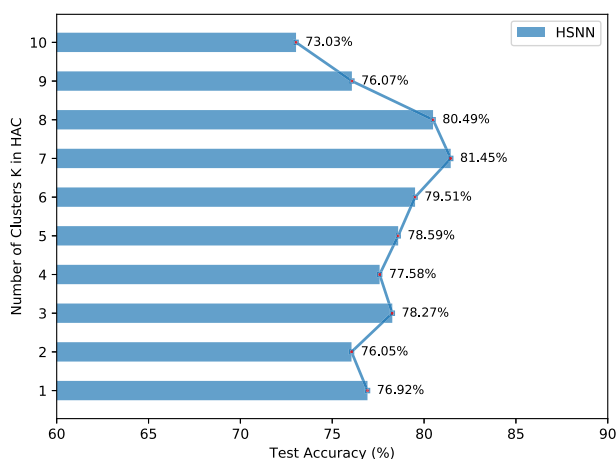


Fig. 7 The performance of HSNN on 20News when varying the number of clusters K

accuracy reaches the peak point, surpassing the baseline by a large margin of 4.5%. After that, the model begins to suffer performance decline. A possible explanation is that the first local classifier is prone to make wrong predictions when the number of clusters increases. The experimental results are in agreement with the results from Experiments I, where our label clustering algorithm determines that the near-optimal number of clusters is 7.

To further prove that the performance boost comes from the exploited label relations rather than from a specific number of clusters in hierarchical classification, we perform ablation experiments by creating four extra sets of cluster configurations. We randomly assign labels to one of the 7 clusters in each configuration, where label relations are neglected in this way. Fig. 8 shows the results. We empirically show that even though HSNN benefits from hierarchical classification compared with the baseline ($K=1$, baseline), the most remarkable accuracy gains come from the best cluster configuration ($K=7$, HAC). The general observation from the experiments is that our label clustering algorithm successfully exploits label relations, which can be leveraged thoroughly by HSNN to produce superior performance.

4.3 Experiments III: Effects of different granularity levels of text representations on LSNN

In LSNN, given a particular dataset, we pre-train two neural networks to extract features in terms of cluster-specific and label-specific predictions, and feed the resulting text representations into consecutive timesteps in the bidirectional LSTM structure. To evaluate the effects of different granularity level of text representations on LSNN as well as to demonstrate that LSNN can capture the flexible relations in the two concept layers to improve text classification performance, we set up four different sets of inputs. The first set follows exactly the original implementation of LSNN, where both cluster-specific and label-specific text representations are used as inputs. It can be regarded as the baseline. In the second and third sets, the inputs consist of cluster-specific or label-specific text representations respectively. In the fourth set, the granularity levels of text representations are not taken into consideration and we jointly train the feature extractors and other components in LSNN together.

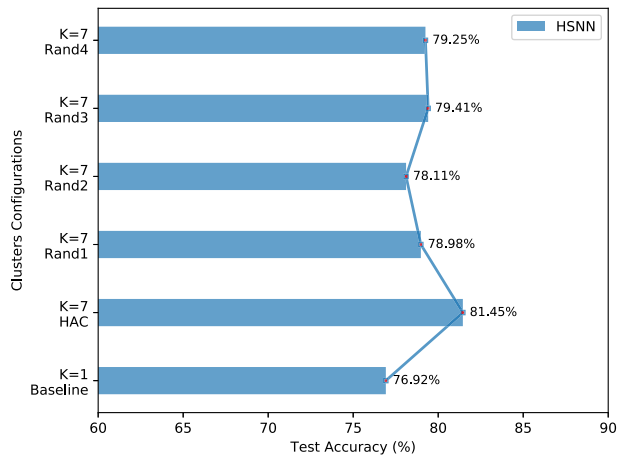


Fig. 8 The performance of HSNN on 20News when assigning all class labels to a specific number of clusters ($K=7$) randomly

We use fastText implemented in Experiments II to extract text representations. The results are reported in Table 2.

We empirically show that when there are only cluster-specific text representations, the model suffers substantial performance decline. A possible reason is that the text representations in terms of cluster-specific predictions is so coarse and discrete that it is difficult to guarantee the model's convergence and consequently the model has trouble discriminating the right class from the rest. Besides, the model trained with label-specific text representations fed into both timesteps yields a slight degradation compared with the baseline. In the baseline, both cluster-specific and label-specific text representations are provided, and LSNN captures label relations implicitly in the consecutive concept layers, which brings moderate accuracy gains.

The experimental settings of the fourth set are different from other sets, where the granularity levels are non-deterministic because they are inferred automatically in the way of joint training. Results show that the performance of LSNN drops compared with the baseline. However, We observe an interesting property in the fourth set. The test accuracy decreases significantly for 20News, but just slightly for DBP and Yah. From the statistics of all three datasets, we can see that the size of 20News is very small compared with others. Provided with a limited number of training examples, the feature extractors can not converge steadily in the joint training, not to mention learning text representations in reasonable granularity levels. The results show the necessity of the pre-train phase in LSNN.

Table 2 Effects of different granularity levels of text representations on LSNN

Dataset	cluster-specific +	cluster-specific +	label-specific +	joint training
	label-specific	cluster-specific	label-specific	
20News.	81.92%	42.70%	78.74%	32.84%
DBP.	98.31%	71.03%	98.02%	97.27%
Yah.A.	72.74%	49.25%	72.37%	70.90%

In conclusion, LSNN benefits from the text representations of different granularity levels from pre-trained feature extractors and it captures label relations implicitly to improve text classification performance.

4.4 Experiments IV: Evaluate the performance of our framework in a comprehensive way

In this section, we report the experimental results of our proposed models leveraging the exploited label relations on multiple text classification datasets. We perform three sets of comparisons here, where in each set a *vanilla* text classification model is chosen as a baseline to deal with all labels in a flat way. The comparisons across different experimental sets are not fair since HSNL uses the vanilla model as the local classifier and LSNN uses it to pre-train feature extractors. In other words, their performances heavily depend on the vanilla model. We have also set up another baseline termed as *o-CRF* which uses a conditional random field over the vanilla model output to leverage the label relations, as shown in [2, 8].

In the first set, our local reproduction of fastText on benchmark datasets seems to replicate the results reported in [14], with a slight variance of less than 0.2%. In the second set, we employ TextCNN [15], one of the most popular convolutional neural networks for text classification. Our implementation of TextCNN has three filter region sizes of 3, 4 and 5, and we use 100 feature maps for each one of them. A fully connected layer of 128 hidden units is placed behind the concatenation of 1-max pooling to extract text representations for LSNN. In the third set, we adopt RCNN [17], which uses a recurrent structure for contextual information and a max-pooling layer for important words. We set the size of the context vector as 64 for both sides, the filter region size in CNN as 3 with 50 feature maps. The experimental results have been shown in Table 3.

The results highlight the fact that our framework is effective for text classification, from exploiting label relations with clustering techniques to performing classification with HSNL or LSNN. Compared with the vanilla model which assumes the semantic independence of labels, the hierarchical classification model HSNL achieves a moderate improvement on accuracy with a group of local classifiers within each experimental set. On the other hand, incorporating the text representations of different granularity levels from pre-trained feature extractors, LSNN also achieves reasonable performance. Besides, both HSNL and LSNN outperform o-CRF in almost all cases. We perform the Friedman test for either HSNL or LSNN against baselines, and the p-value is less than the significance level of 0.05, showing that our models perform significantly better.

Among all datasets, we observe that HSNL and LSNN produce more performance gains on 20News compared with the other two datasets. We attribute this observation to the fact that some labels in 20News share a more coherent semantic space, which has been demonstrated in Fig. 5. Consequently, label relations can be exploited and leveraged more thoroughly, and they contribute to more improvements for text classification. On DBP, the performance improvement seems to be not so significant, because the original performance achieved by the vanilla model is so high that there is less room for notable improvement. In some extreme cases where label relations are weak (e.g., in Fig. 5, K=3 for Yah shows the lowest average silhouette value among all the optimal K), the performance gains will be marginal.

Table 3 Test accuracy in three sets of comparisons

Model	FastText				RCNN				TextCNN			
	Vanilla	o-CRF	HSNN	LSNN	Vanilla	o-CRF	HSNN	LSNN	Vanilla	o-CRF	HSNN	LSNN
20News.	76.92%	78.12%	81.45%	81.92%	74.58%	74.81%	76.30%	78.56%	75.53%	76.12%	77.62%	79.21%
DBP.	97.92%	98.02%	98.13%	98.31%	98.14%	97.95%	98.24%	98.42%	97.88%	98.05%	97.98%	98.23%
Yah.A.	72.03%	72.24%	72.61%	72.74%	72.11%	72.30%	72.88%	72.81%	71.65%	72.01%	72.31%	73.04%

5 Conclusion

In this paper, we present a label clustering algorithm to exploit the flexible label relations. We adopt HAC with the silhouette value for the partition of disjoint clusters to generate the two-layer label space. The exploited label relations are further leveraged in our proposed neural networks, HSNN and LSNN, for the task of text classification. HSNN is inspired by the idea of hierarchical classification and consists of multiple local classifiers in the form of a cascaded tree structure. LSNN receives text representations from pre-trained feature extractors as inputs and leverages label relations in the way of bidirectional inferences. We have performed extensive experiments to evaluate the effectiveness of our proposed framework. The results show that both HSNN and LSNN leverage the exploited label relations effectively to generate better text classification results.

In the future, we hope to extend our model to exploit a deeper hierarchy path of label relations while preventing the propagation of classification errors to improve its performance further. Other future research directions include putting forward new algorithms for label clustering or designing novel deep learning models to make use of label relations better.

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